

Prediction and correlation

Agenda

- + Today:
 - + Statistical communication exercise
 - + prediction and correlation
- + Reminders:
 - + Project Part 1 due Friday
 - + Don't wait too late to try loading data into R!
 - + Detailed instructions on the data exploration cheat sheet (course website)
 - + See me if you have any issues

Statistical communication exercise

You are a statistical consultant, and a client reaches out to you to ask about some data they have collected. You fit a linear model, and the client has a question for you.

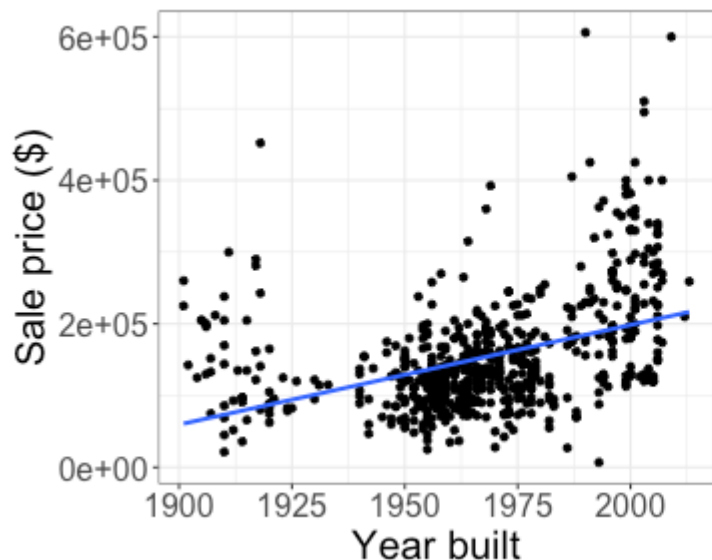
Spend ~5 minutes, individually, writing a brief email to the client to answer their question. Then we will discuss as a class.

Last time: SSE

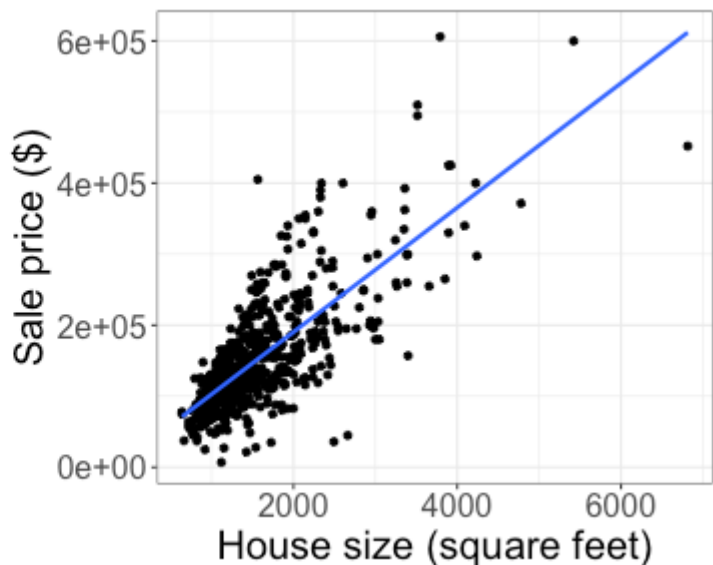
Sum $\searrow$ shared residuals

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

residual



SSE: 3.41×10^{12}



SSE: 1.74×10^{12}

These numbers are very large!

Last time: SSE

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- + Depends on the scale of the response variable
- + Depends on the *number* of observations
- + Depends on the strength of the relationship

Alternative: RMSE

← root mean square error

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- + Depends on the scale of the response variable
- + Depends on the *number* of observations
- + Depends on the strength of the relationship

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

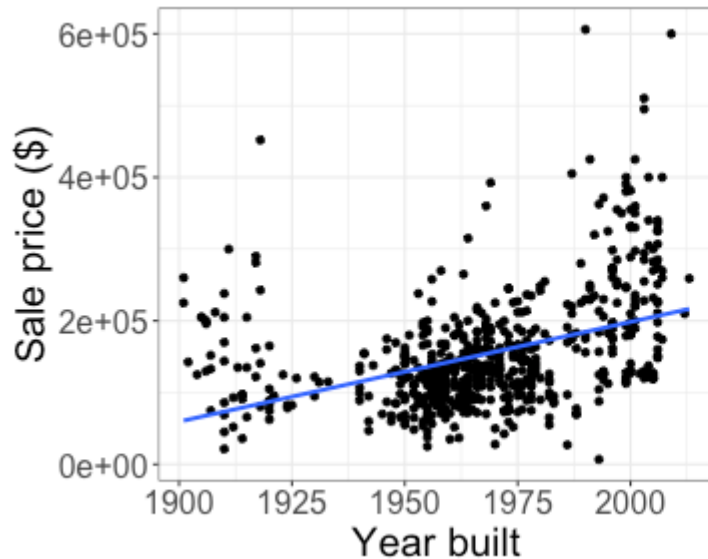
↑ divide by # of observations

- + Depends on the scale of the response variable
- + Depends on the strength of the relationship
- + Does *not* depend on the number of observations

square root: RMSE is on same scale as original data

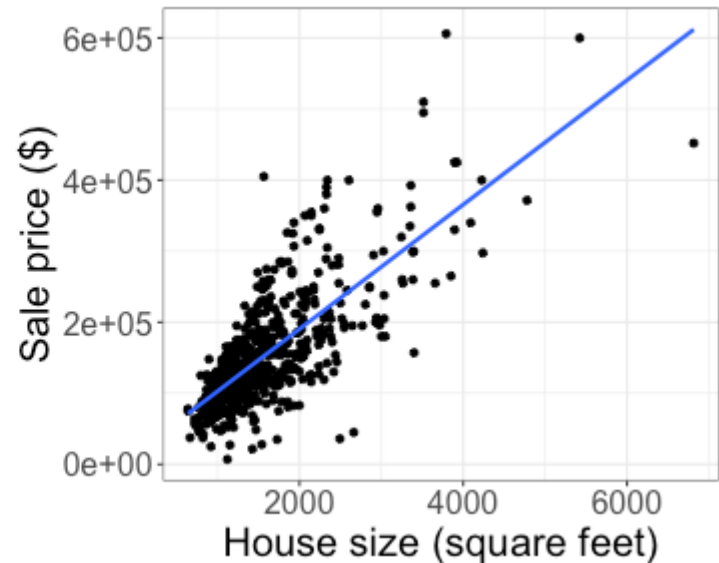
Root mean square error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



RMSE: 73121

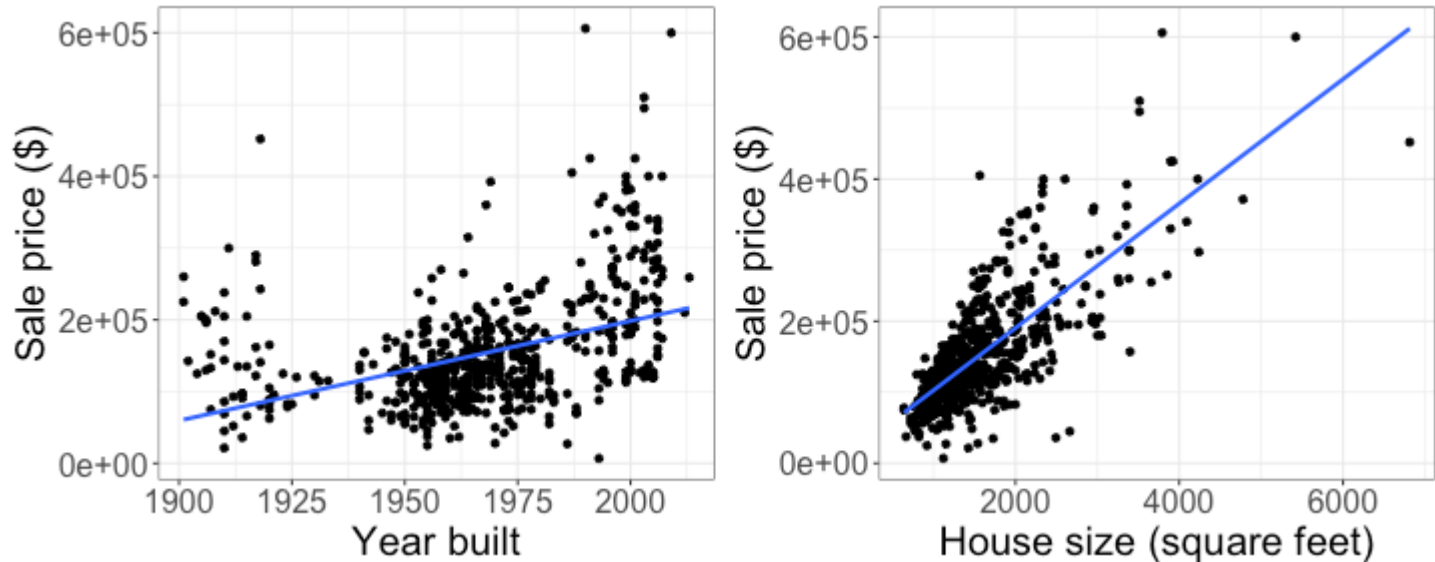
on average, our predicted house price is about \$73K away from the true house price



RMSE: 52745

on average, our predicted house price is about \$53K away from the true house price

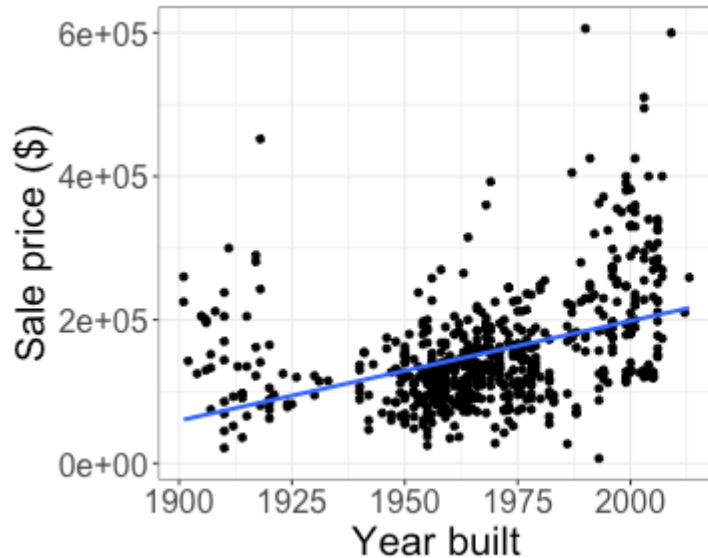
Root mean square error (RMSE)



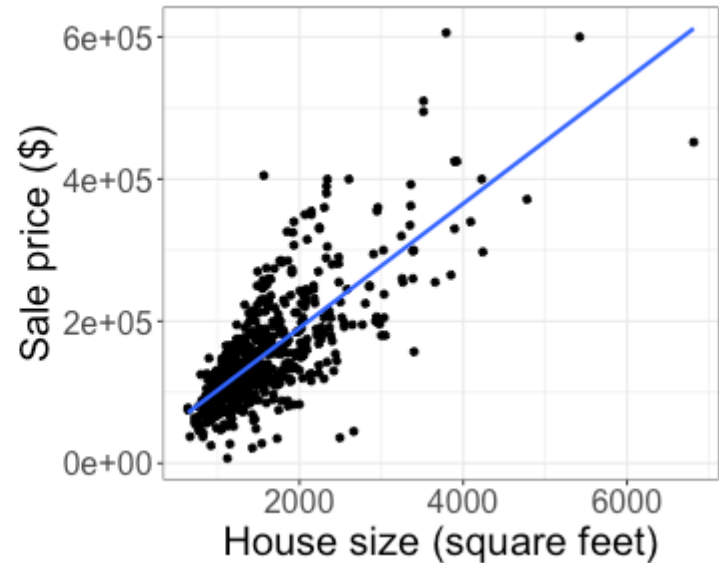
- + Both SSE and RMSE are useful for summarizing our ability to make predictions
- + Both depend on the scale of the response variable

Is there a measure of the strength of a linear relationship that does *not* depend on scale?

Correlation



Correlation: 0.41



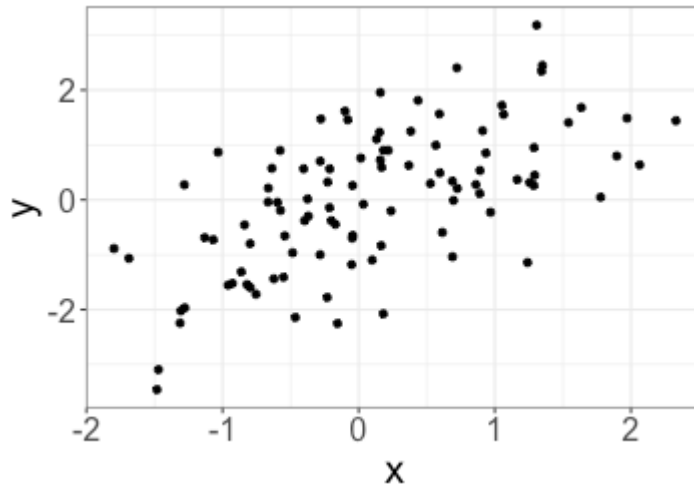
Correlation: 0.75

What is the possible range of values for a correlation?

between -1 and 1

Correlation

Observe points $(x_1, y_1), \dots, (x_n, y_n)$



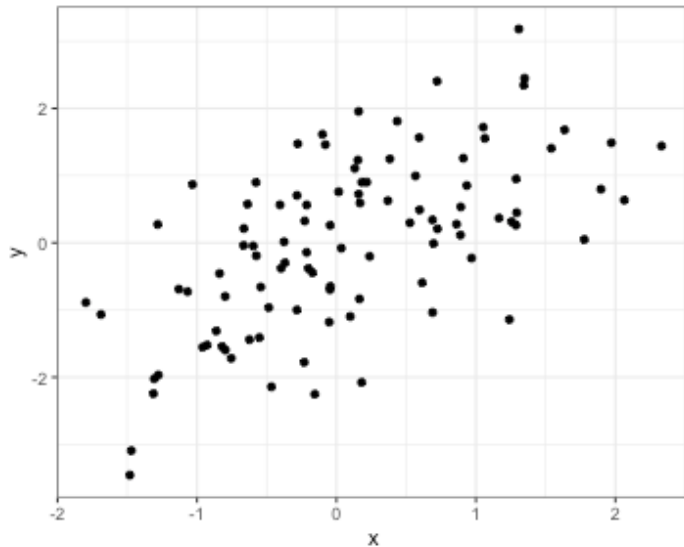
$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

how closely
do the x &
 y points go together

variability in x

variability in y

Correlation



(A) 0.2

(B) -0.2

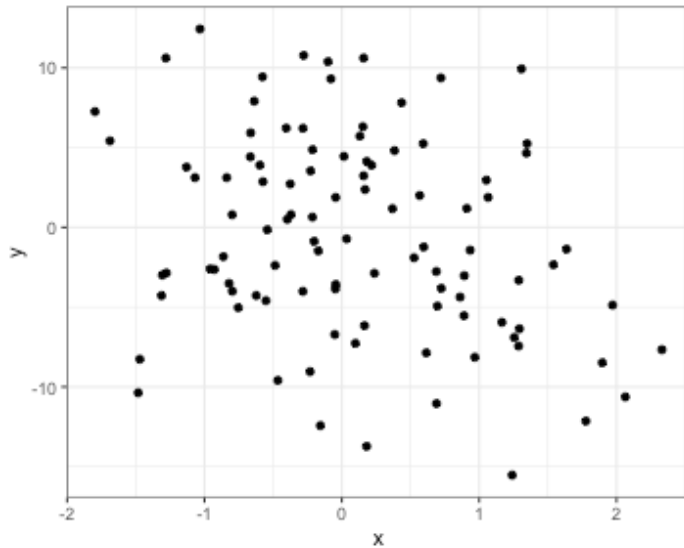
(C) 0.6

*moderate, positive
correlation*

(D) 0.9

What would you estimate as the correlation for this data?

Correlation



(A) 0

(B) -0.28

(C) 0.28

(D) -0.6

What would you estimate as the correlation for this data?

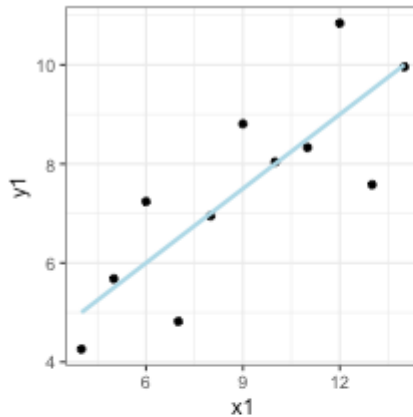
Correlation

Key takeaway:

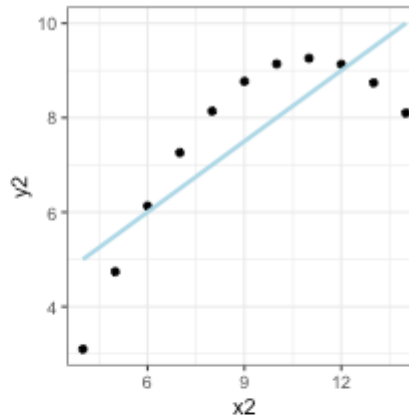
correlation doesn't tell us if the line is a good choice, just measures how close points are to the line

Which of these plots shows the highest correlation?

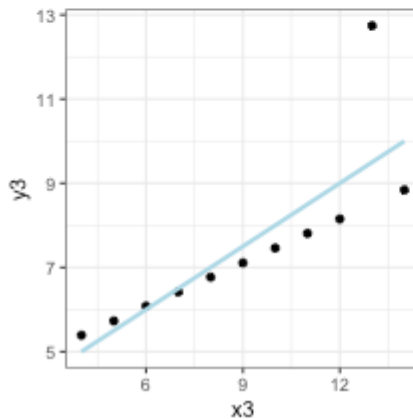
$$r = 0.816$$



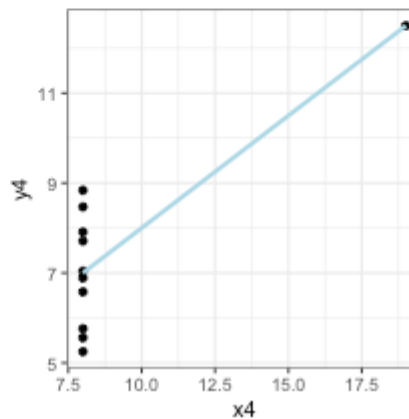
$$r = 0.816$$



$$r = 0.816$$



$$r = 0.816$$



Prediction and correlation

- + SSE:

- + depends on strength of the relationship
- + depends on number of observations
- + depends on scale of the response

- + RMSE:

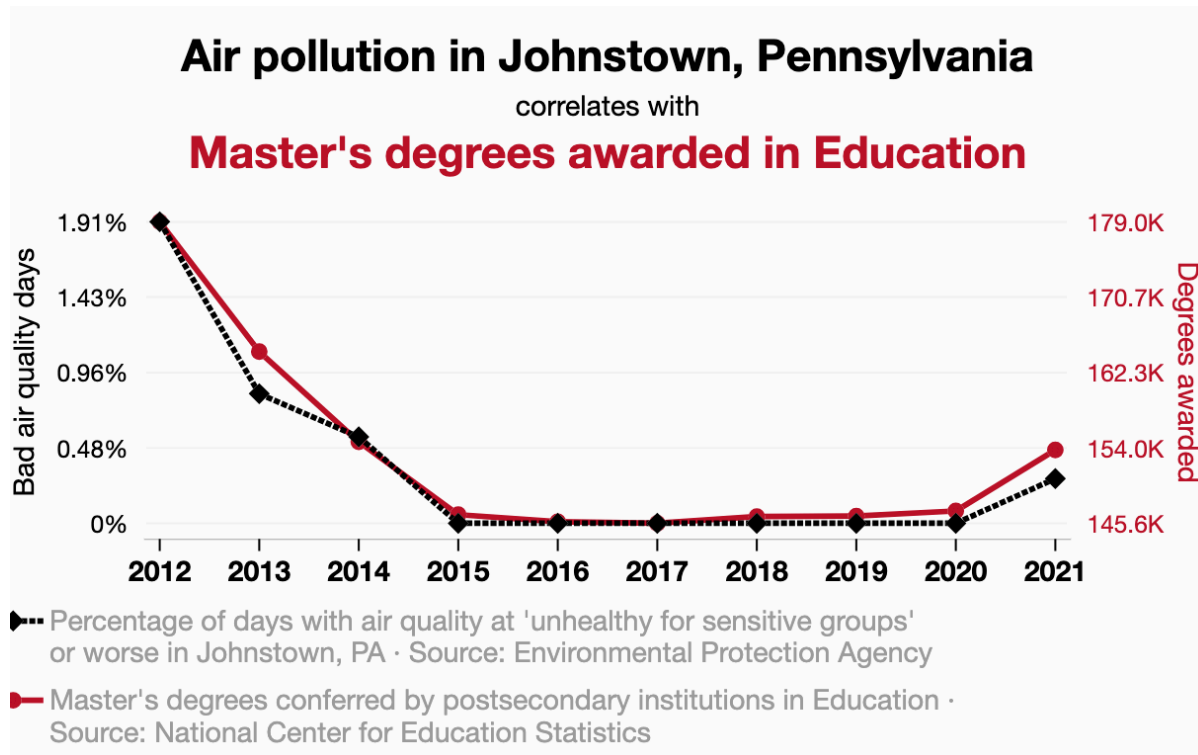
- + depends on strength of the relationship
- + depends on scale of the response

- + Correlation:

- + depends on strength of the relationship
- + values always between $[-1, 1]$
- + measures strength of a **linear** relationship -- may not be suitable for nonlinear relationships
- + influenced by outliers and extreme points

Correlation?

What is going on in this plot? Do you think this correlation is meaningful?



$r = 0.989$

Activity: spurious correlations

https://sta112-f25.github.io/class_activities/ca_07.html

- + Play around with spurious correlations!
- + No need to submit anything for this activity